Binary Coded Decimal

BCD (1248)

=> Also called positively weighted code
=> BCD is a way to express each of the decimal digits with a binary code.
=> There are only 10 valid codes with 6 invalid codes as given below.

Decimal digit	BCD (8421)
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

These are valid

These are Invalid.....

10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

Decimal to BCD

To convert any decimal number to BCD, simply replace each decimal digit with a valid 4-bit code.

Example:
Convert each of the following decimal numbers to BCD.

(a) 35

$\underbrace{0011}_{3}$ $\underbrace{0101}_{5}$

$(35)_{10} = (00110101)_{BCD}$

(b) 98

9 → 1001

8 → 1000

$(98)_{10} = (10011000)_{BCD}$

(c) 170

1 → 0001

7 → 0111

0 → 0000

$(170)_{8} = (000101110000)_{BCD}$

(d) 2469

2 → 0010

4 → 0100

6 → 0110

9 → 1001

$(2469)_{10} = (0010010001101001)$

BCD to Decimal

Example: Convert each of the following BCD to Decimal.

(a) 10000110

1000 0110

8 6

$(86)_{10}$ = 1000110

(d) 110010

0011 0010

3 2 = $(32)_{10}$

(b) 001101010001

$= (351)_{10}$

(c) 1001000

0100 1000 $= (48)_{10}$

BCD Addition

Rules:

1) Perform addition like binary addition.

ii) if sum ≤ 9, then it is a valid BCD number

iii) if sum > 9 or a carry a carry out of a 4-bit group is generated, then an invalid result a generated.

iv) Add 6(0110) to the 4-bit sum to skip the six invalid states.

v) if a carry results when 6 is added, simply add the carry to the next 4-bit.

Example: Add the following BCD numbers: (valid)

(a) 0011 + 0100

0011	3
0100	+4
0111	7

(b) 00100011 + 00010101

0010 0011	23
+ 0001 0101	+ 15
0011 1000	38

(c) 10000110 + 00010011

1000 0110	86
+ 0001 0011	+ 13
1001 1001	99

Example: Add the following BCD Numbers (Invalid).

(a) 1001 + 0100

1001	9
+ 0100	+ 4
1101	13

Invalid BCD numbers >9 ← 1101

+0110

$\boxed{1}$ 0011

0001 0011

$\boxed{00010011}$ valid BCD number

(b) 1001 + 1001 = ?

1001

+ 1001

← $\boxed{1}$ 0010

0110

0001 1000

$\boxed{00011000}$ Valid

(c) 00010110 + 00010101

0001 0110

+ 0001 0101

0010 1011 → invalid >9

1 0110

0011 $\boxed{1}$ 0001

(00110001) Ans

(d)

01100111 + 01010011

0110 0111

+ 0101 0011

1011 1010

+ 0110 0110

← $\boxed{1}$ 0001 $\boxed{1}$ 0000

Both groups are invalid add >9 6(0110)

0001 0001

1

0010

$\boxed{0001\ 0010\ 0000}$ Ans

Note ≡ BCD is a self complementing code. Self complementing codes provide the 9's complement of a decimal number, just by interchanging 1's and 0's in its equivalent (2421 representation

Add 1001 + 1000

1 0 0 1	9
+ 1 0 0 0	+8
$\boxed{1}$ 0 0 0 1	17
0 1 1 0	
0 1 1 1	

0001

00010111 **Ans**

Add 1001 + 0111

1 0 0 1	9
+ 0 1 1 1	+7
$\boxed{1}$ 0 0 0 0	16
0 1 1 0	
0 1 1 0	

0001 0110 **Ans**

1 6

(e)

00100101 + 00100111

0 0 1 0	0 1 0 1	25
+ 0 0 1 0	0 1 1 1	+27
0 1 0 0	1 1 0 0	52 → Inva…
1	0 1 1 0	29 a…
0 1 0 1	$\boxed{1}$ 0 0 1 0	

0101 0010 Ans

5 2

Gray Code

Also called unit distance code.

⇒ The gray code is unweighted and non arithmetic code.

⇒ It exhibits only a single bit change from one code word to the next in sequence.

Decimal	Binary	Gray code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

from above adjacent numbers it is clear that there is only a single bit change in Gray coded number to the next in sequence. However binary number change appears of more than one bit often.

Binary to-Gray code Conversion

Rules:

(i) The most significant bit (left most) in the Gray code is the same as the MSB in the binary number.

(ii) Going from left to the right, added each adjacent pair of binary code bits to get the next Gray coded bit.

(iii) Discard all the carries produced.

Example:-

Convert the binary number 10110 to Gray code.

MSB

(Binary) $\boxed{1} \xrightarrow{+} 0 \xrightarrow{+} 1 \xrightarrow{+} 1 \xrightarrow{+} 0$

Remains Same ← $\boxed{1}$ 1 1 0 1

↓ Discard carry

So 10110 = (11101) (in Gray code)

Convert the binary number 11000110 to Gray code.

1 1 0 0 0 1 1 0

$\boxed{1}$ 0 1 0 0 1 0 1

So 11000110 = (0100101 (Gray code)

Gray to binary Code:

Rules:

(i) The most significant bit in the binary code is the same as the MSB in the Gray code.

(ii) Add each binary code generated to the gray code bit in the next adjacent position.

(iii) Discard all the carries produced.

Example: 10101111 to binary

(Gray) 1 0 1 0 1 1 1 1

1 1 0 0 1 0 1 0

So

Graycode 10101111 = (11001010 (in Binary code)